

Templates : Template/PG-fr.docx | Template/Presentation_BELACEL_V3.pptx

PDF : TP_Informatique_ENS_1ère_année_AN_01.pdf

Instructor: BELACEL Madani

Module: Computer Science

Level: ENS — Year 1 — English Department

Duration: 1h30

Academic Year: 2025-2026

Learning Objectives

At the end of this lab, the student is able to:

- Install and configure Python on their machine
- Write and run a first Python script
- Use `print()` and `input()` functions
- Declare and manipulate variables (int, float, str, bool)
- Perform arithmetic operations and unit conversions
- Structure a simple program with conditions

Lab Plan

Phase	Content	Duration
Welcome	Python installation & editor	15 min
Exercise 1	First script	10 min
Exercise 2	Variables and types	15 min
Exercise 3	Arithmetic operations	15 min
Exercise 4	Unit converter	20 min
Exercise 5	Average calculator	15 min
Exercise 6	Challenge — Pedagogical calculator	15 min

Prerequisites

- Python 3 installed (check with `python --version`)
- A text editor (VS Code, IDLE, Thonny, or even Notepad)

Exercise 1 — First script (10 min)

Problem statement: Create a Python script named `presentation.py` that displays your last name, first name, and group.

Expected output:

Last name : BELACEL
First name : Madani
Group : ENS1 English

Solution:

```
print("Last name : BELACEL")  
print("First name : Madani")  
print("Group : ENS1 English")
```

Exercise 2 — Variables and types (15 min)

Problem statement: Create a script `variables.py` that:

1. Declares a variable `name` (str), `age` (int), `height` (float), `is_student` (bool)
2. Displays the type of each variable with `type()`
3. Displays a summary sentence using an f-string

Expected output:

```
<class 'str'>  
<class 'int'>  
<class 'float'>  
<class 'bool'>  
My name is Madani, I am 20 years old, I am 1.75 m tall, student : True
```

Solution:

```
name = "Madani"  
age = 20  
height = 1.75  
is_student = True  
  
print(type(name))  
print(type(age))  
print(type(height))  
print(type(is_student))  
  
print(f"My name is {name}, I am {age} years old, I am {height} m tall, student :  
{is_student}")
```

Exercise 3 — Arithmetic operations (15 min)

Problem statement: Create a script `calculs.py` that:

1. Asks the user for two numbers with `input()`
2. Calculates and displays their sum, difference, product, quotient, and power
3. Handles the case where the second number is 0 (display "Division impossible")

Expected output:

```
First number : 10  
Second number : 3  
10 + 3 = 13  
10 - 3 = 7  
10 × 3 = 30  
10 / 3 = 3.33  
10 ^ 3 = 1000
```

Solution:

```

a = float(input("First number : "))
b = float(input("Second number : "))

print(f"{a} + {b} = {a + b}")
print(f"{a} - {b} = {a - b}")
print(f"{a} × {b} = {a * b}")

if b != 0:
    print(f"{a} / {b} = {a / b:.2f}")
else:
    print("Division impossible")

print(f"{a} ^ {b} = {a ** b}")

```

Exercise 4 — Unit converter (20 min)

Problem statement: Create a script `convertisseur.py` that:

1. Asks for a temperature in degrees Celsius
2. Converts to Fahrenheit ($F = C \times 9/5 + 32$)
3. Converts to Kelvin ($K = C + 273.15$)
4. Displays the results with 2 decimal places

Expected output:

```

Temperature in °C : 25
25.00°C = 77.00°F
25.00°C = 298.15 K

```

Solution:

```

celsius = float(input("Temperature in °C : "))
fahrenheit = celsius * 9/5 + 32
kelvin = celsius + 273.15

print(f"{celsius:.2f}°C = {fahrenheit:.2f}°F")
print(f"{celsius:.2f}°C = {kelvin:.2f} K")

```

Converter flowchart:

```

flowchart TD
    A[Start] --> B[Enter temperature in °C]
    B --> C{Valid input?}
    C -->|No| D[Display error]
    D --> B
    C -->|Yes| E[Calculate Fahrenheit]
    E --> F[Calculate Kelvin]
    F --> G[Display results]
    G --> H[End]

```

Exercise 5 — Average calculator (15 min)

Problem statement: Create a script `moyenne.py` that:

1. Asks the user for 3 grades
2. Calculates the average
3. Displays the average with 2 decimal places
4. Displays "Admitted" if average ≥ 10 , "Failed" otherwise

Expected output:

```
Grade 1 : 14
Grade 2 : 12
Grade 3 : 16
Average : 14.00/20 – Admitted
```

Solution:

```
n1 = float(input("Grade 1 : "))
n2 = float(input("Grade 2 : "))
n3 = float(input("Grade 3 : "))

average = (n1 + n2 + n3) / 3

print(f"Average : {average:.2f}/20", end=" – ")
if average >= 10:
    print("Admitted")
else:
    print("Failed")
```

Exercise 6 — Challenge: pedagogical calculator (15 min)

Problem statement: Create a script `calculatrice.py` that:

1. Asks the user for two numbers
2. Asks for the desired operation (+, -, *, /)
3. Performs the operation and displays the result
4. Displays an error message if the operation is unknown or if division by zero

Expected output:

```
Number 1 : 15
Number 2 : 4
Operation (+, -, *, /) : /
15.0 / 4.0 = 3.75
```

Solution:

```
a = float(input("Number 1 : "))
b = float(input("Number 2 : "))
op = input("Operation (+, -, *, /) : ")

if op == "+":
    print(f"{a} + {b} = {a + b}")
elif op == "-":
    print(f"{a} - {b} = {a - b}")
elif op == "*":
    print(f"{a} × {b} = {a * b}")
elif op == "/":
    if b != 0:
        print(f"{a} / {b} = {a / b}")
    else:
        print("Error: division by zero")
else:
    print("Unknown operation")
```

Calculator flowchart:

```
flowchart TD
    A[Start] --> B[Enter number a]
    B --> C[Enter number b]
```

```

C --> D[Enter operator]
D --> E{Operator?}
E --> "+" | F[Display a + b]
E --> "-" | G[Display a - b]
E --> "*" | H[Display a × b]
E --> "/" | I{b = 0 ?}
I --> Yes | J[Error division by zero]
I --> No | K[Display a / b]
E --> Other | L[Unknown operation]
F --> M[End]
G --> M
H --> M
J --> M
K --> M
L --> M

```

Self-assessment quiz

1. Which function displays a message on the screen?

- a) input()
- b) print() ✓
- c) type()

2. What is the result of type(3.14)?

- a) <class 'int'>
- b) <class 'float'> ✓
- c) <class 'str'>

3. What is the value of 17 % 5 in Python?

- a) 3
- b) 2 ✓
- c) 3.4

4. Which conversion is correct for inputting a decimal number?

- a) int(input("Grade : "))
- b) float(input("Grade : ")) ✓
- c) str(input("Grade : "))

5. In an f-string, how do you insert the variable age?

- a) {age} ✓
- b) [age]
- c) (age)

Grading Rubric Lab 01

Criterion	Weight	Not acquired (0)	Partial (0.5)	Acquired (1)
Installation and execution	15%	Cannot execute	Executes with help	Executes independently

Basic Python syntax	20%	Blocking errors	Some errors	Correct code
Variables and types	20%	Does not declare variables	Partially correct types	Correct types
Operations and I/O	20%	Does not handle inputs	Partially correct operations	Everything works
Converter and average	15%	Not completed	Partially completed	Completed
Calculator challenge	10%	Not completed	Partially functional	Fully functional

Session schedule — 1h30 (90 min)

Phase	Duration	Cumulative total
Welcome and installation	15 min	15 min
Exercise 1 — First script	10 min	25 min
Exercise 2 — Variables and types	15 min	40 min
Exercise 3 — Arithmetic operations	15 min	55 min
Exercise 4 — Unit converter	20 min	75 min
Exercise 5 — Average calculator	15 min	90 min
Exercise 6 — Calculator challenge	—	At home (bonus)

Pedagogical Note ENS

Teaching transposition

This introductory Python lab is designed for future teachers of French who will teach computer science in middle and high school (SNT in Seconde, NSI in general high school). The pedagogical progression follows increasing difficulty: discovery → manipulation → creation. The unit converter (exercise 4) is an excellent tool for addressing magnitudes and measurements in interdisciplinary work with physical sciences. The calculator challenge (exercise 6) can be offered as a home project and constitutes a reusable model in class: a middle school student can customize it to create their own pedagogical calculator.

Trainer advice

Ensure that each student has Python installed and working before the start of the lab. Allow 15 minutes of buffer time for installation issues. Encourage pair work for mutual assistance but require individual submissions. For middle school students, replace the calculator challenge with a program that displays "Hello + first name" to avoid cognitive overload.

Extension

Students can create a complete "Digital grade book" program that records daily lessons, exercises to do, and assessments, with the date and grade level concerned.